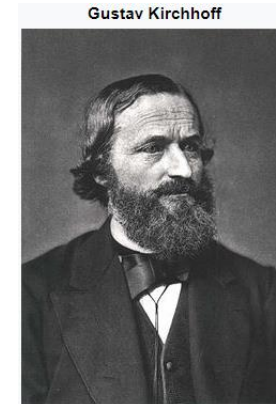
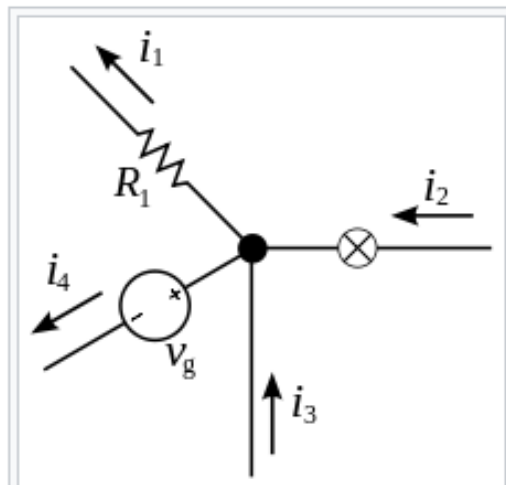


Ch27 Circuits

Gustav Robert Kirchhoff (1824 –1887) was a German physicist who contributed to electrical circuits, spectroscopy, and black-body radiation.

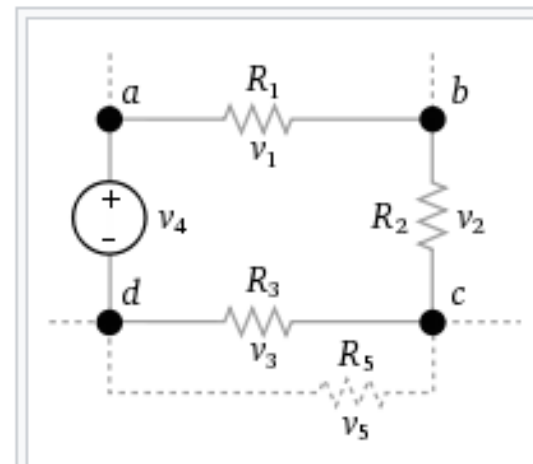


Kirchhoff's current law (KCL)



The current entering any junction is equal to the current leaving that junction. $i_2 + i_3 = i_1 + i_4$

Kirchhoff's voltage law (KVL)



The sum of all the voltages around a loop is equal to zero.
 $v_1 + v_2 + v_3 - v_4 = 0$

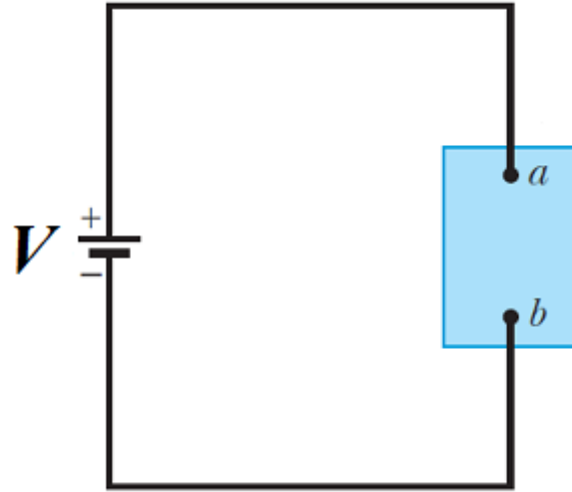
Today's topics (Chapter 27)

■ Emf — ξ

■ Kirchhoff Loop Rules

■ RC Circuit

Circuits

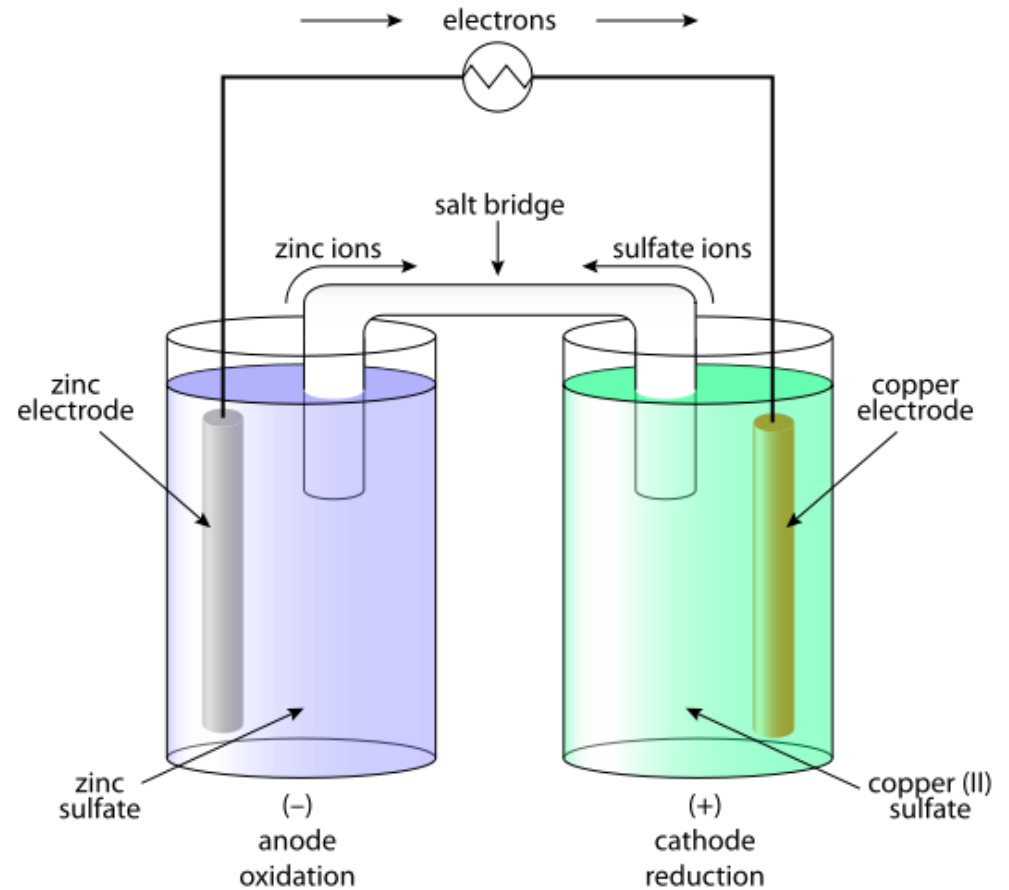
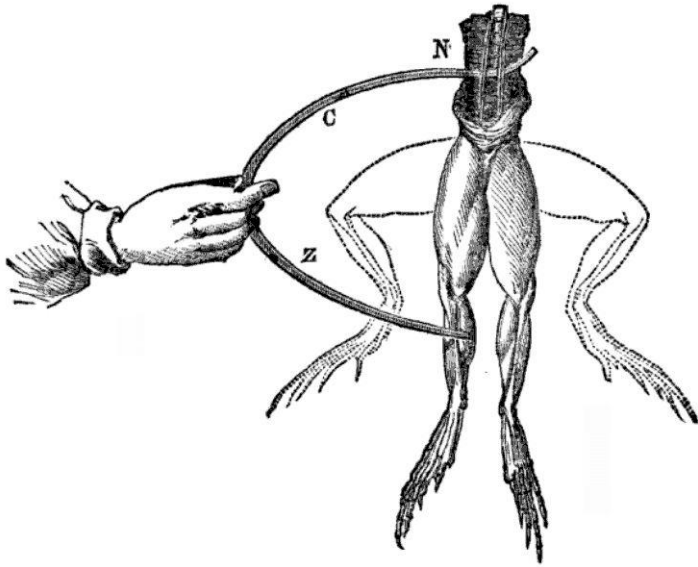


The influence that makes current flow from lower to higher potential is called electromotive force **emf ξ**

An ideal emf maintains a constant potential difference its terminals, independent of the current through it.

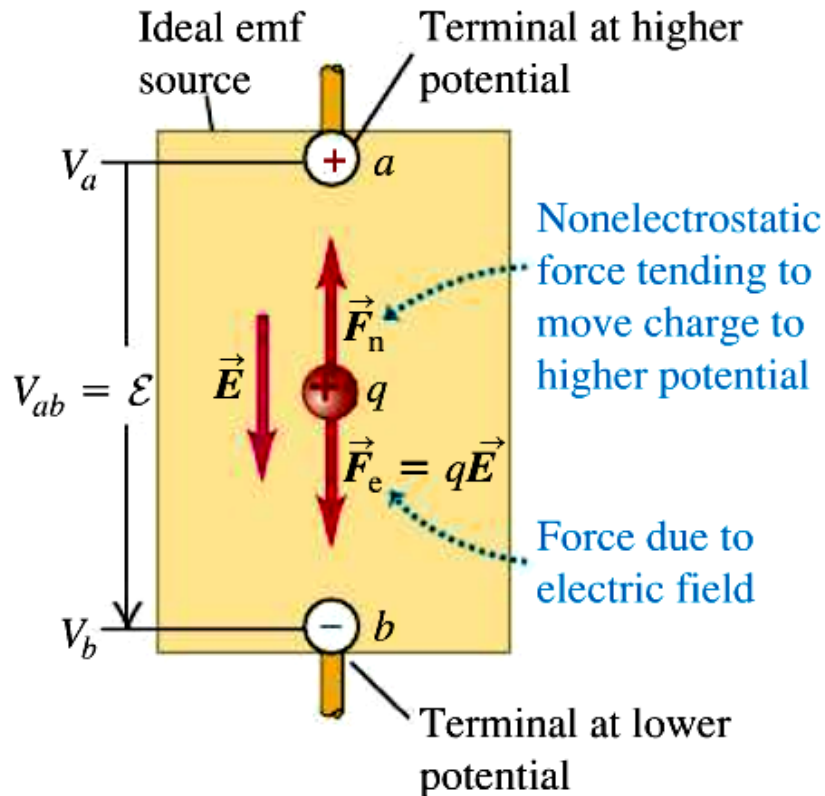
$$V_{ab} = \mathcal{E} \quad (\text{ideal source of emf})$$

Galvanic cell



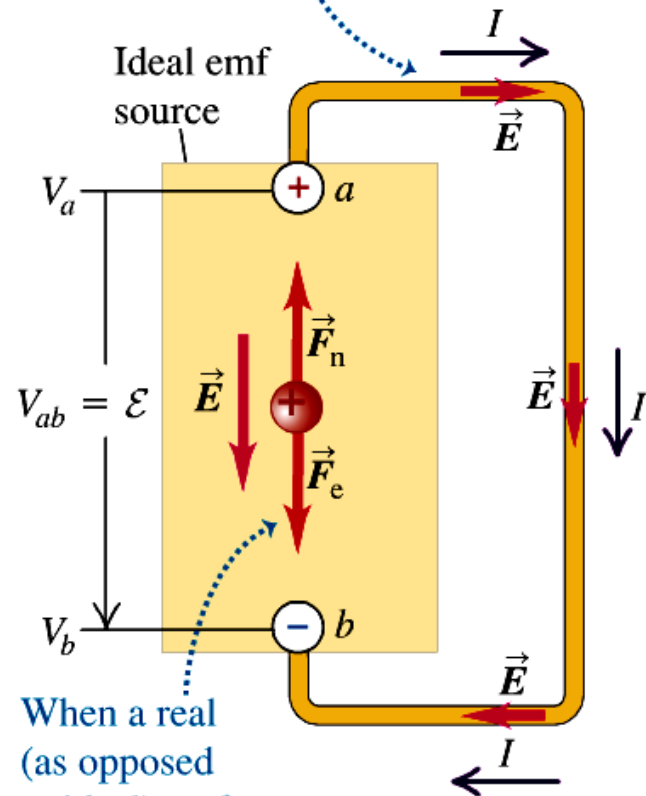
https://en.wikipedia.org/wiki/Luigi_Galvani
https://en.wikipedia.org/wiki/Galvanic_cell

How does a battery work?



When the emf source is not part of a closed circuit, $F_n = F_e$ and there is no net motion of charge between the terminals.

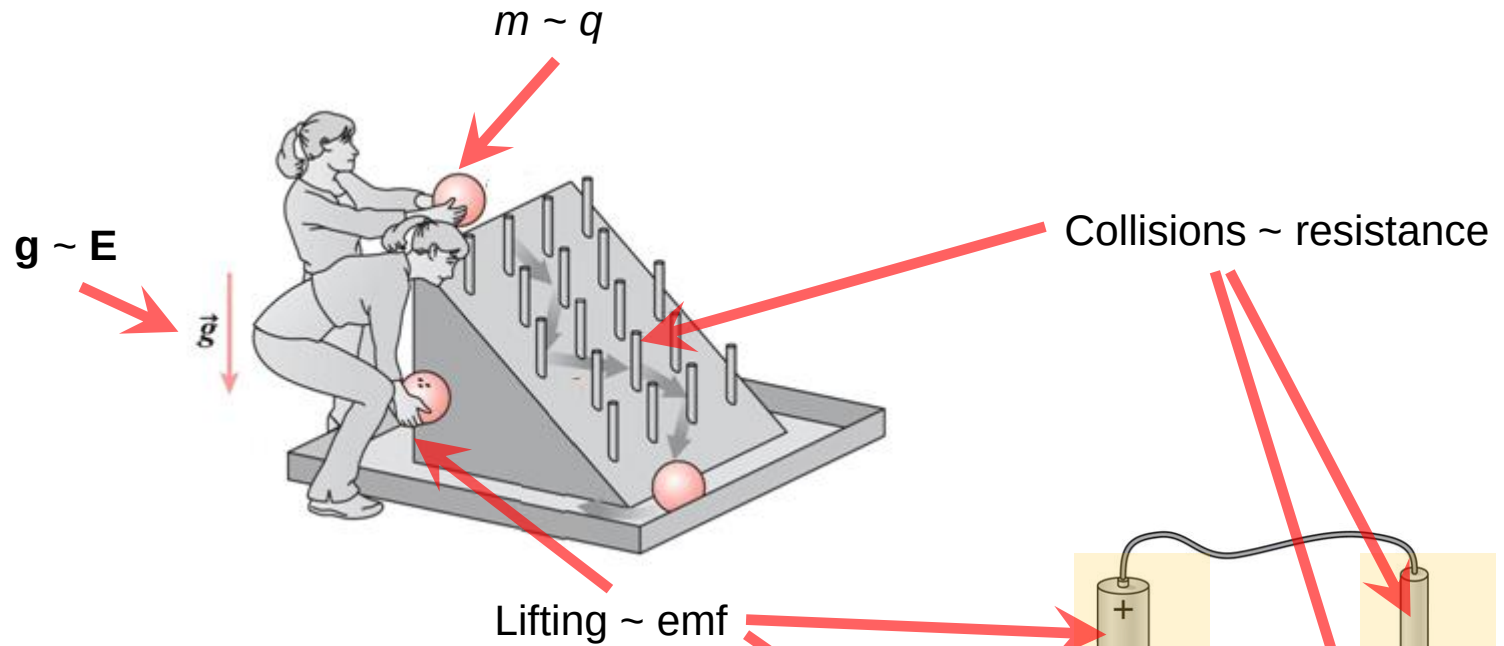
Potential across terminals creates electric field in circuit, causing charges to move.



When a real (as opposed to ideal) emf source is connected to a circuit, V_{ab} and thus F_e fall, so that $F_n > F_e$ and $\vec{F}_n$ does work on the charges.

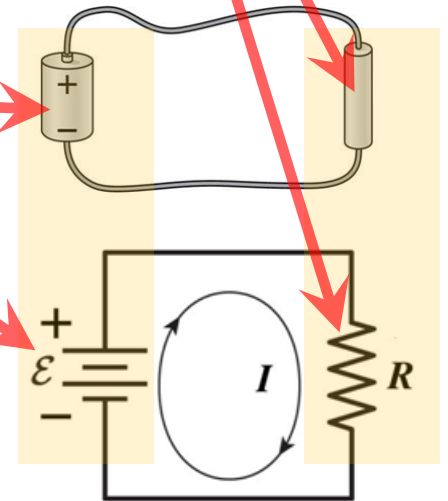
<http://electronics.howstuffworks.com/everyday-tech/battery.htm>

Battery is like Sisyphus



Energy gained by charge transversing
battery = $q e$ (To be dissipated as heat in
external R.)

Ideal emf : no internal energy loss.



Ohm's law:
$$I = \frac{\mathcal{E}}{R}$$

Internal resistance in a real source of *emf*

In an ideal *emf* : $V_{ab} = \mathcal{E}$



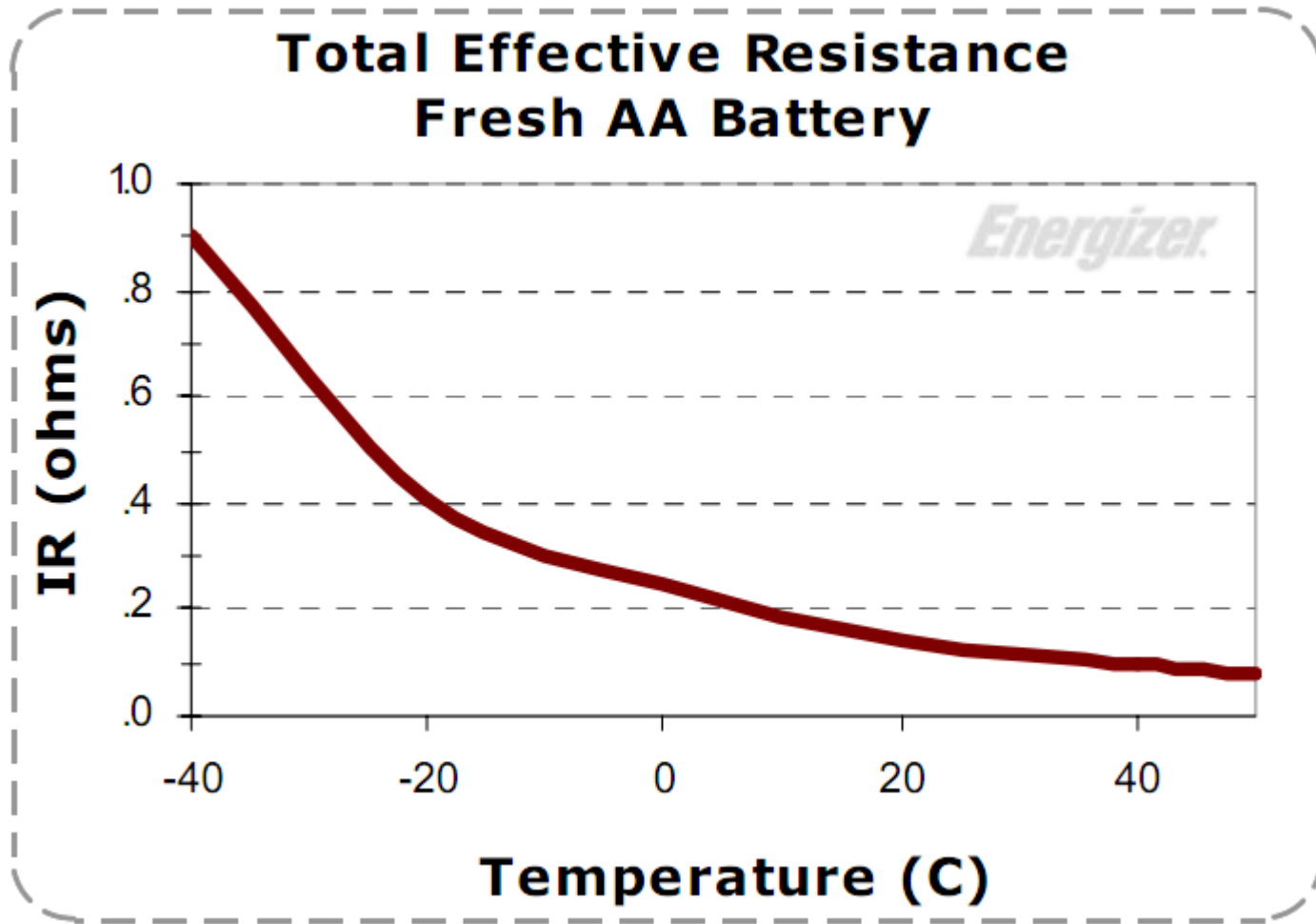
1.5 V ?

In a real (non-ideal) *emf* : $V_{ab} = \mathcal{E} - Ir$



For a real source of *emf*, terminal voltage equals the *emf* only if no current is flowing through the source.

Internal resistance VS temperature for battery



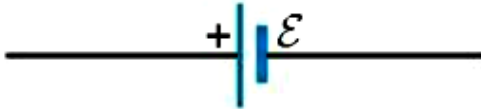
Symbols for circuit diagram



Conductor with negligible resistance



Resistor

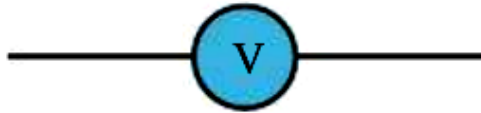


Source of emf (longer vertical line always represents the positive terminal, usually the terminal with higher potential)

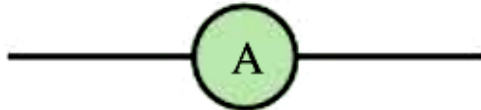


Source of emf with internal resistance r (r can be placed on either side)

or



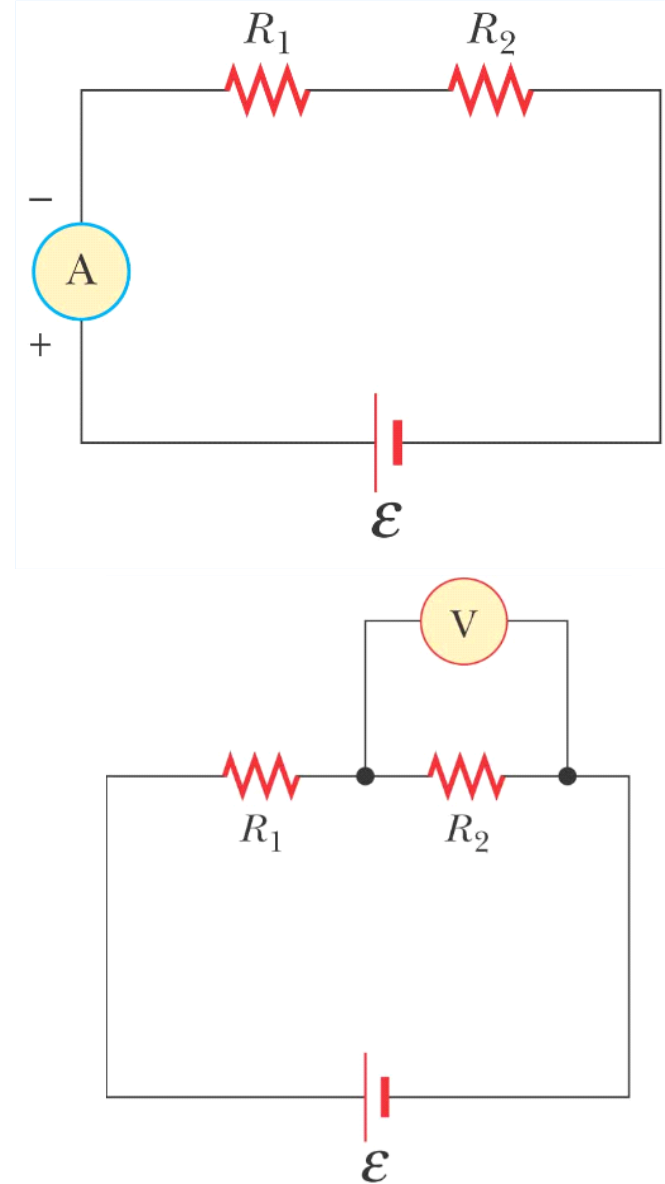
Voltmeter (measures potential difference between its terminals)



Ammeter (measures current through it)

Ammeter and Voltmeter

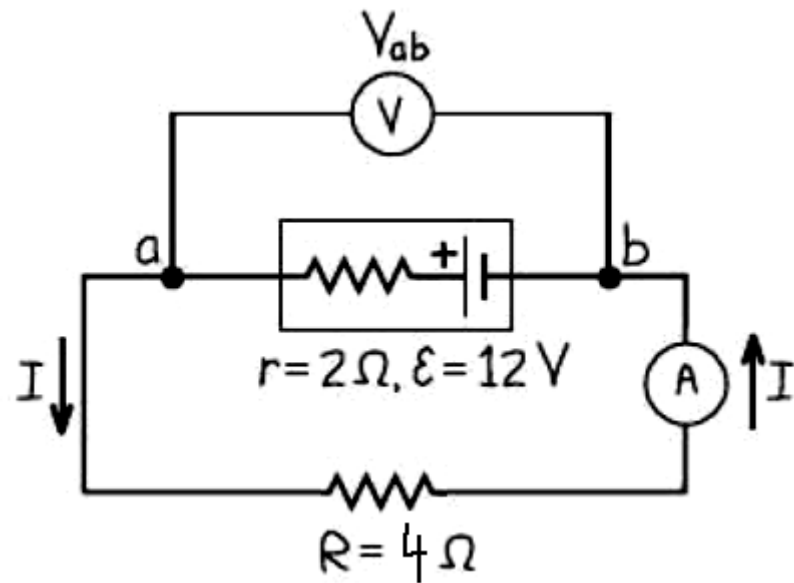
- Ammeter – measures current
- Connects in series with measured elements in circuit
- Ideal ammeter should have zero resistance so that current being measured is not altered
- Voltmeter – measures potential difference
- Connects between two points in a circuit
- Ideal voltmeter has infinite resistance so that no current exists in it



A source with a short circuit

$$V_{ab} = \mathcal{E} - Ir = 0 \quad \text{If } R = 0$$

$$I = \frac{\mathcal{E}}{r} = \frac{12 \text{ V}}{2 \Omega} = 6 \text{ A}$$



$$\text{If } R = 4 \Omega$$

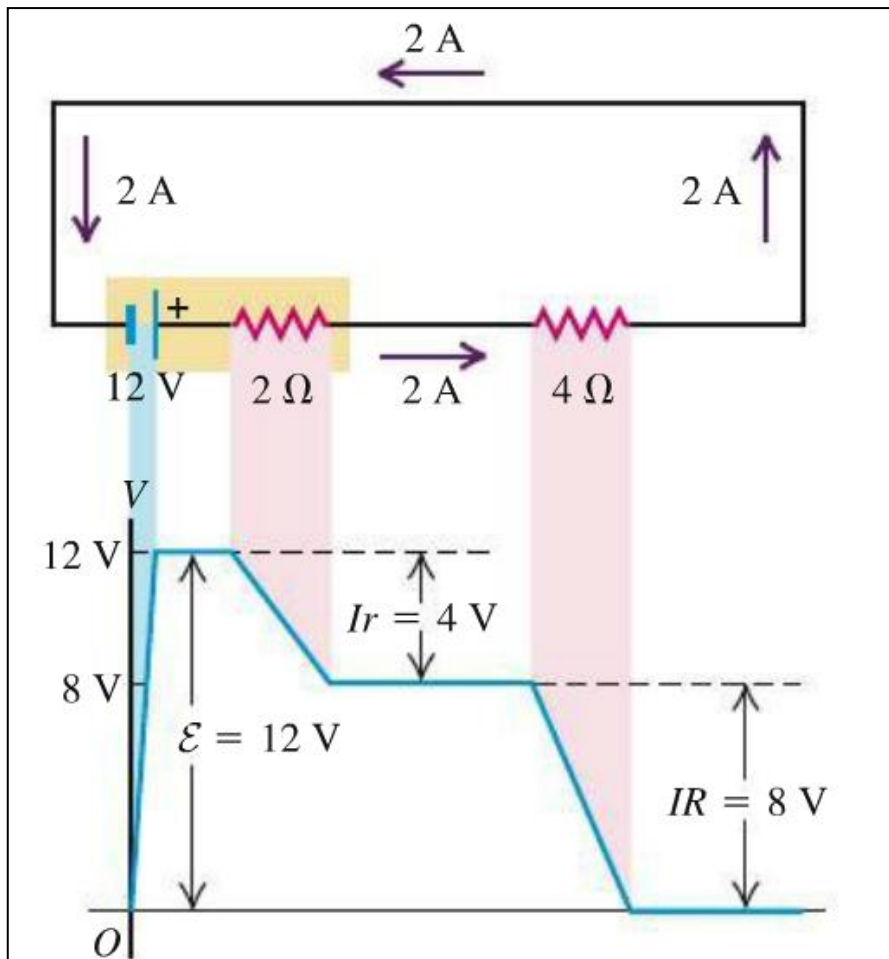
$$\mathcal{E} - Ir - IR = 0$$

Power deliver to the circuit:

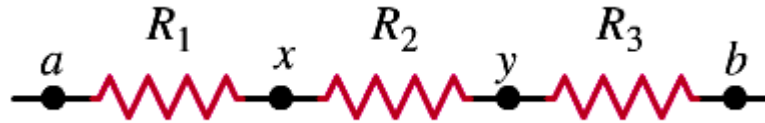
$$\mathcal{E}I - I^2r = I^2R$$

Terminal Voltage
of the battery = 8 V

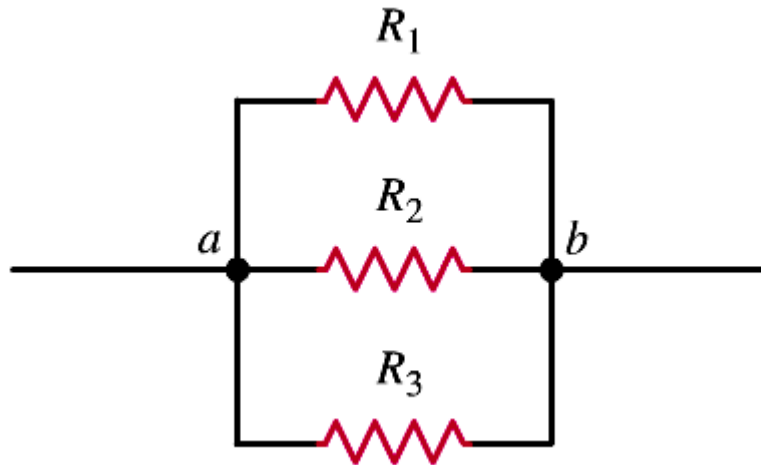
↑
Rate of energy generated by
the chemical reaction



Resistors combination



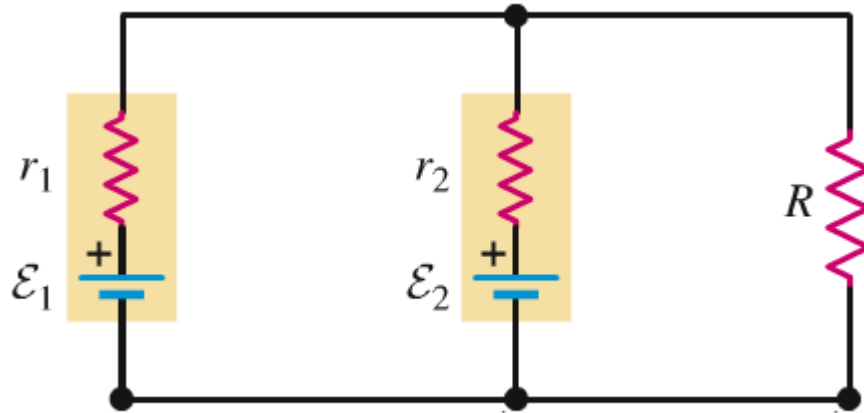
$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots \quad (\text{resistors in series})$$



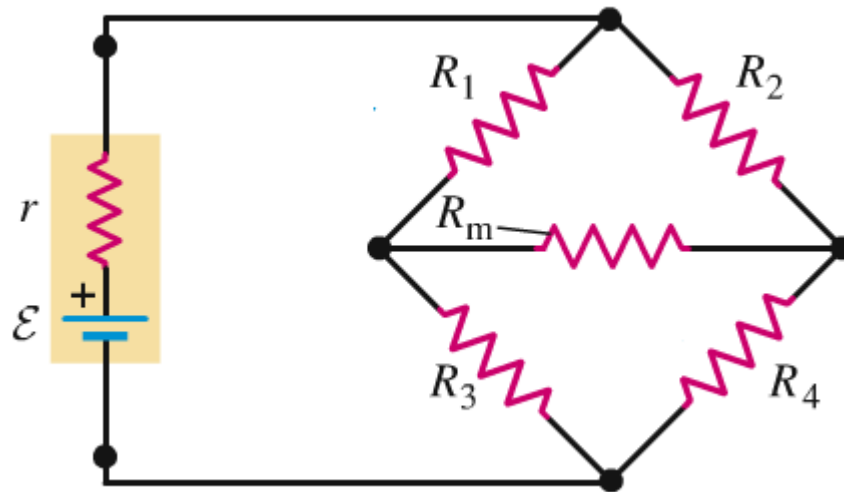
$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad (\text{resistors in parallel})$$

More complicated circuits

e.g.



e.g.



Today's topics (Chapter 27)

- Emf

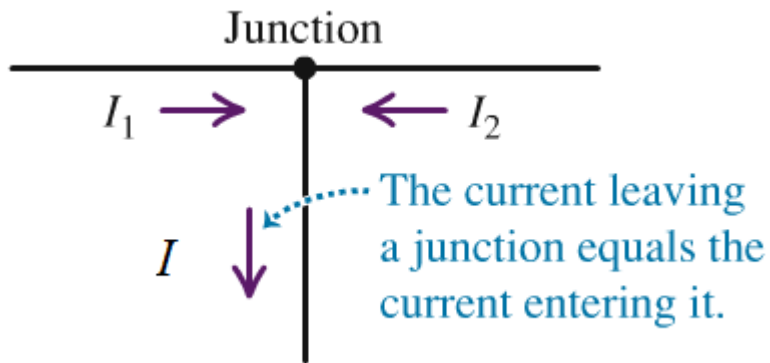
- Kirchhoff Loop Rules

- RC Circuit

Kirchhoff's laws

1) Kirchhoff's junction rule:

The algebraic sum of the currents into any junction is zero, i.e. **at any junction** : $\sum I = 0$



$$I_1 + I_2 + (-I) = 0$$

$$\text{i.e. } I_1 + I_2 = I$$

$\therefore$ charge is conserved

How to determine the sign of the currents?

Kirchhoff's rules

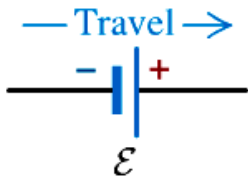
2) Kirchhoff's loop rule:

The algebraic sum of the potential difference of **any loop** is zero, including those associated with emfs or resistive elements, i.e. $\sum V = 0$

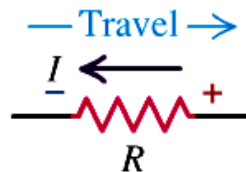
First, we choose a direction, either clockwise or counter-clockwise

From **lower potential to higher potential** $\rightarrow +v$

$+\mathcal{E}$: Travel direction
from $-$ to $+$:

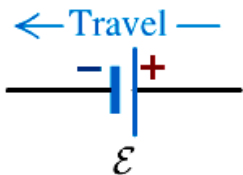


$+IR$: Travel *opposite*
to current direction:

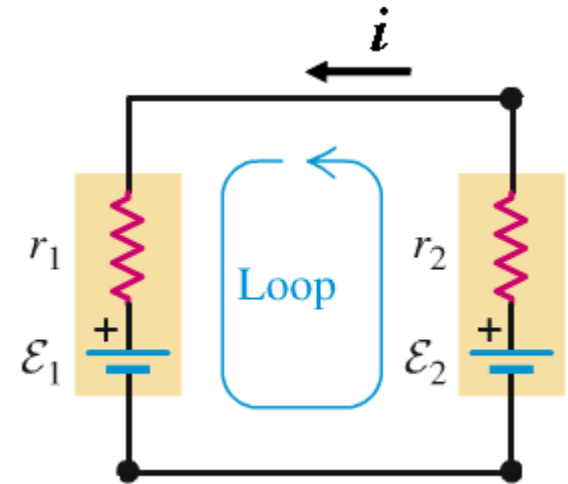
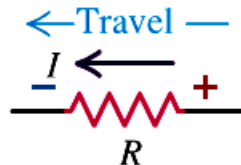


From **higher potential to lower potential** $\rightarrow -v$

$-\mathcal{E}$: Travel direction
from $+$ to $-$:



$-IR$: Travel *in*
current direction:

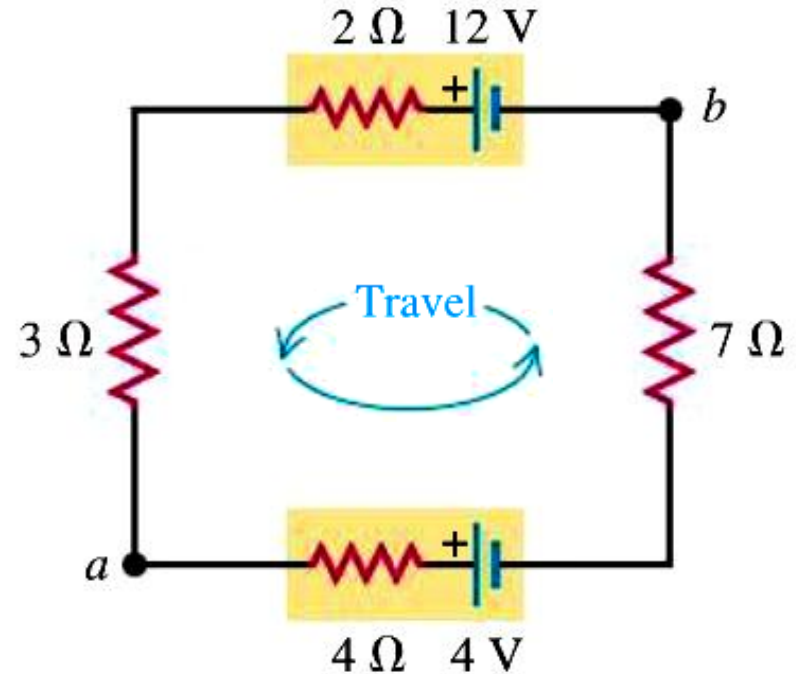


$$\mathcal{E}_2 - ir_2 - ir_1 - \mathcal{E}_1 = 0$$

Example : Find I , V_{ab} , and *power of the emfs*

(a) I

$$\begin{aligned} -I(4\Omega) - 4V - I(7\Omega) + 12V \\ -I(2\Omega) - I(3\Omega) &= 0 \\ 8V &= I(16\Omega) \\ I &= 0.5\text{ A} \end{aligned}$$



(b) V_{ab} Start at b , go to a

$$\begin{aligned} V_{ab} &= 12V - (0.5A)(2\Omega) - (0.5A)(3\Omega) \\ &= 9.5\text{ V} \end{aligned}$$

high to low $\rightarrow -v$
low to high $\rightarrow +v$

What if we take the lower one ?

$$\begin{aligned} V_{ab} &= (0.5A)(7\Omega) + 4V + (0.5A)(4\Omega) \\ &= 9.5\text{ V} \text{ same as above!} \end{aligned}$$

What if we had chosen
a clockwise loop?

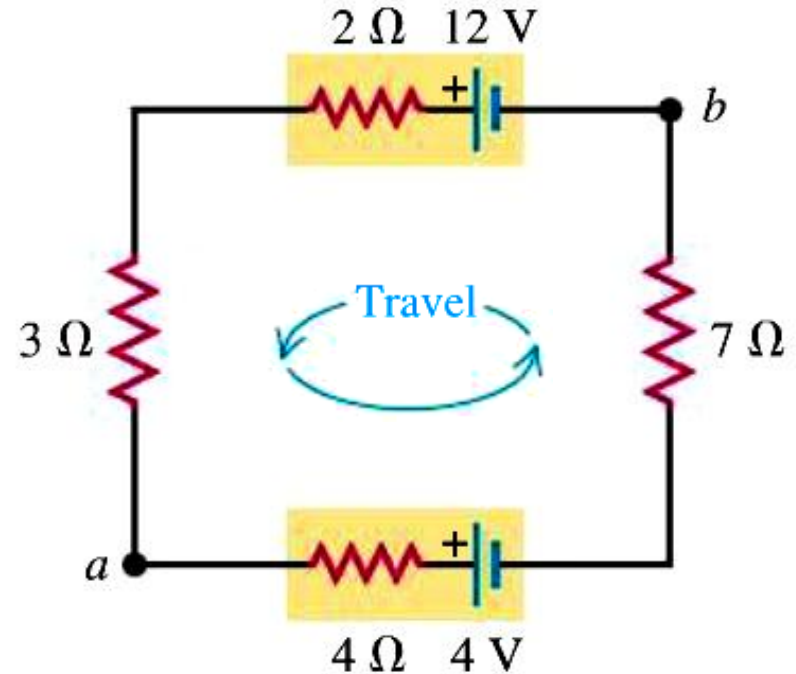
Example : Find I , V_{ab} , and *power of the emfs*

(c) *Power of the emfs*

$$\begin{aligned} P_{12V} &= \varepsilon I = (12V)(0.5A) \\ &= 6 \text{ W} \end{aligned}$$

$$\begin{aligned} P_{4V} &= \varepsilon I = (-4V)(0.5A) \\ &= -2 \text{ W} \end{aligned}$$

-v, we are storing energy to it (charging)!

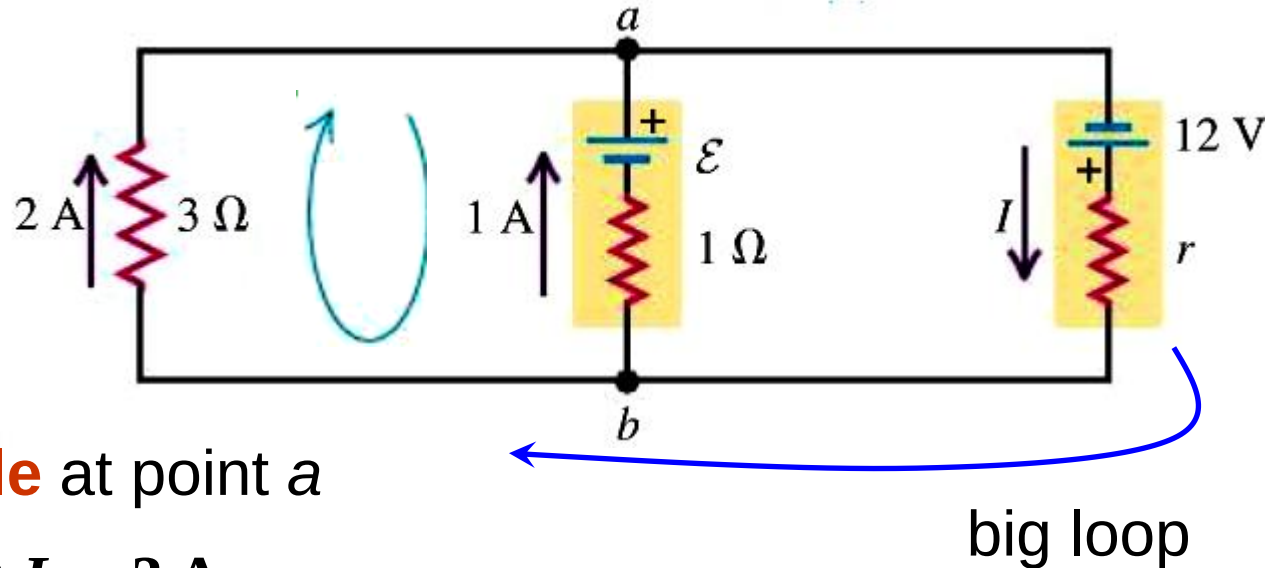


high to low $\rightarrow -v$
low to high $\rightarrow +v$

Example : Find r , ε and current I

3 unknowns

We need three equations!



1) Apply **junction rule** at point a

$$1\text{ A} + 2\text{ A} = I \quad \therefore I = 3\text{ A}$$

2) Apply **loop rule**, use the big loop

$$12\text{ V} - (3\text{ A})r - (2\text{ A})(3\Omega) = 0 \quad \therefore r = 2\Omega$$

3) Apply **loop rule**, use the left loop

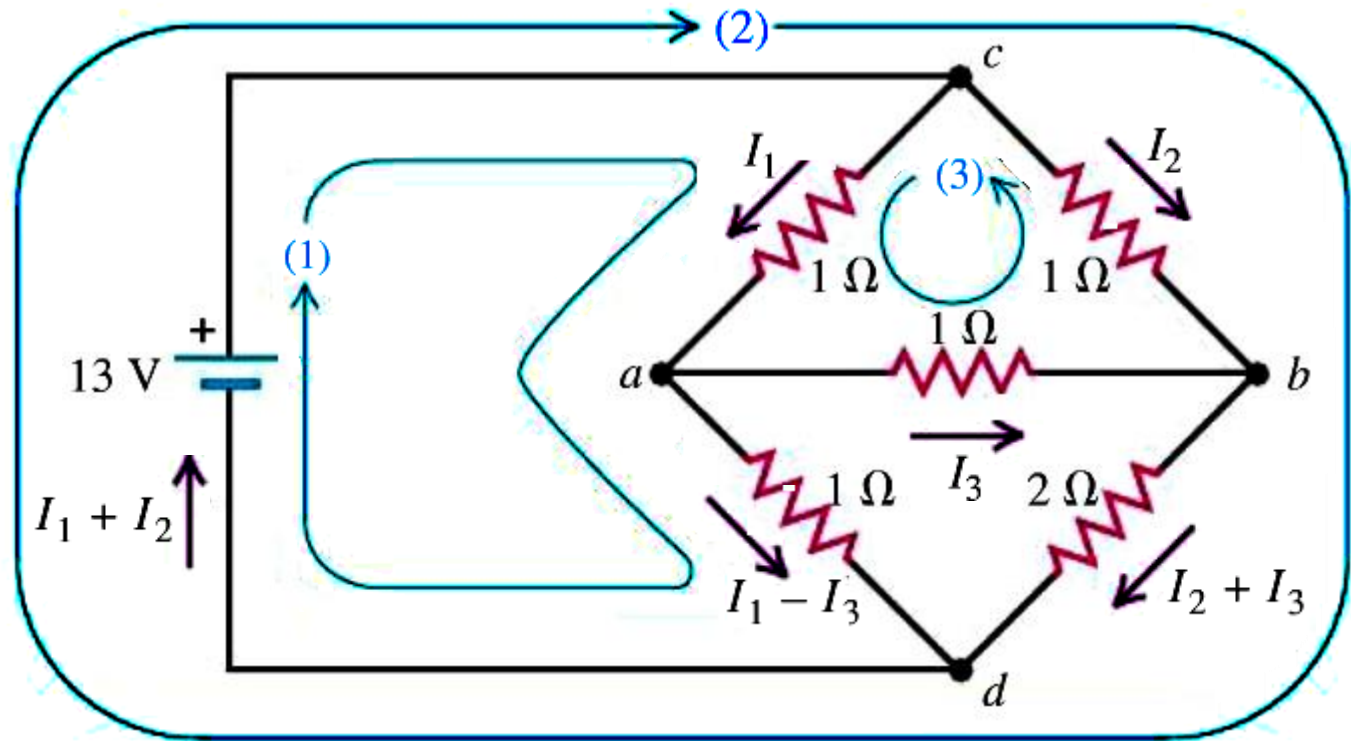
$$-\varepsilon + (1\text{ A})(1\Omega) - (2\text{ A})(3\Omega) = 0 \quad \therefore \varepsilon = -5\text{ V}$$

What does the minus sign mean?

Use junction at point b or the right loop will give the same result!

A complex network

high to low $\rightarrow -v$
low to high $\rightarrow +v$



$$[1] \quad 13\text{V} - I_1(1\Omega) - (I_1 - I_3)(1\Omega) = 0$$

$$[2] \quad -I_2(1\Omega) - (I_2 + I_3)(2\Omega) + 13\text{V} = 0$$

$$[3] \quad -I_1(1\Omega) - I_3(1\Omega) + I_2(1\Omega) = 0$$

Solve the 3 equations for I_1 , I_2
and I_3 , we have :

$$I_1 = 6\text{A}, \quad I_2 = 5\text{A}, \quad I_3 = -1\text{A}$$

How many loops are there in the circuit? Why don't we use all of them?

A problem likely to appear in your quizzes...

All resistor is $4\ \Omega$

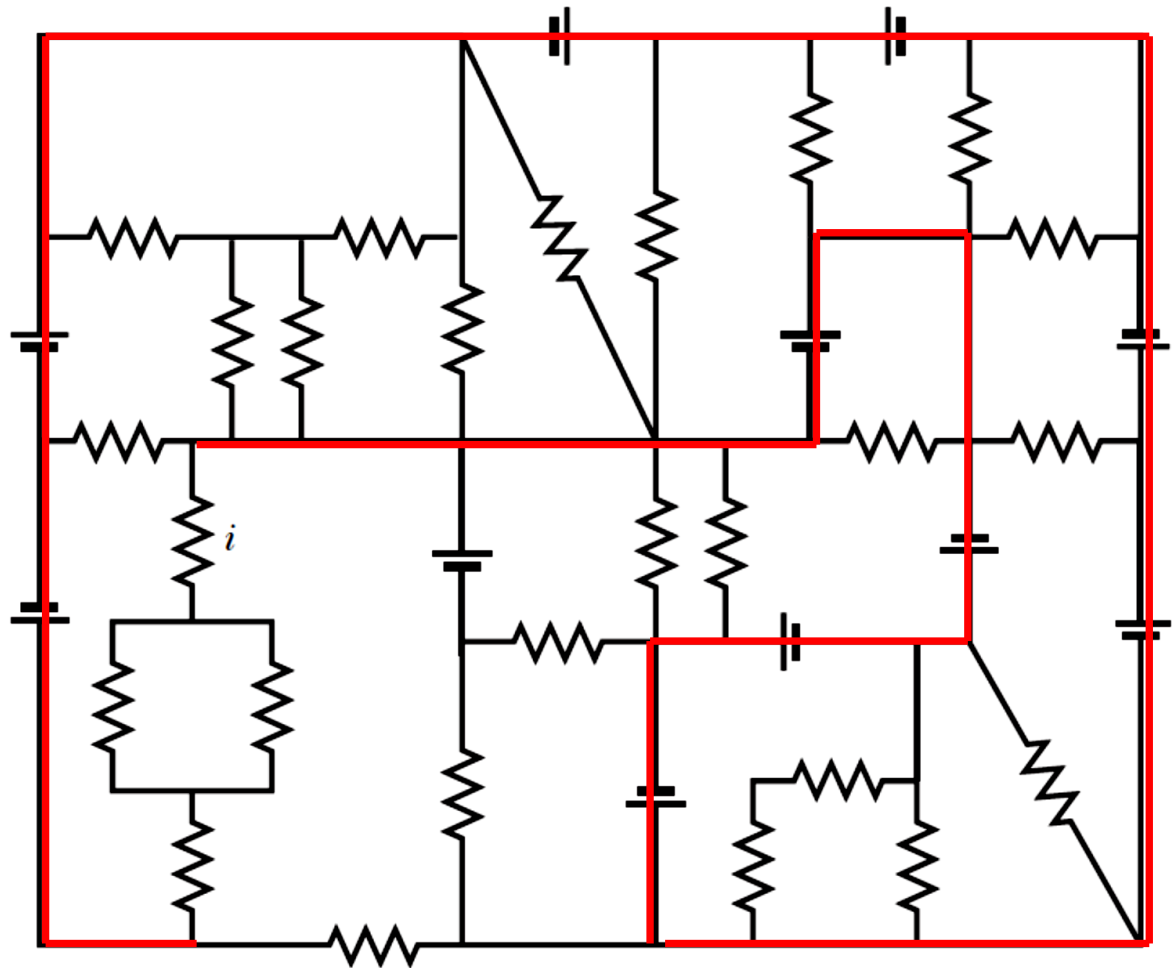
All batteries is 12 V

What is *i* ?

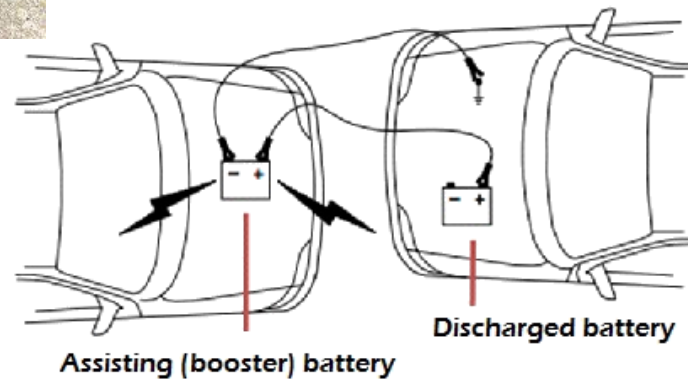
$$R = 10 \, \Omega,$$

$V = 12 \text{ V} \times 4$ going up

$I = 4.8 \text{ A}$ going up



Jump starting a car



Jump starting a car

Weak battery only

$$I = V/R = (10.1 \text{ V})/(0.25 \Omega) = 40 \text{ A}$$

When connected to a good battery
outside loop gives

$$12.5 \text{ V} - I_1(2R_J + r_1) - I_3 R_S = 0$$

$$12.5 \text{ V} - I_1(0.025 \Omega) - I_3(0.15 \Omega) = 0$$

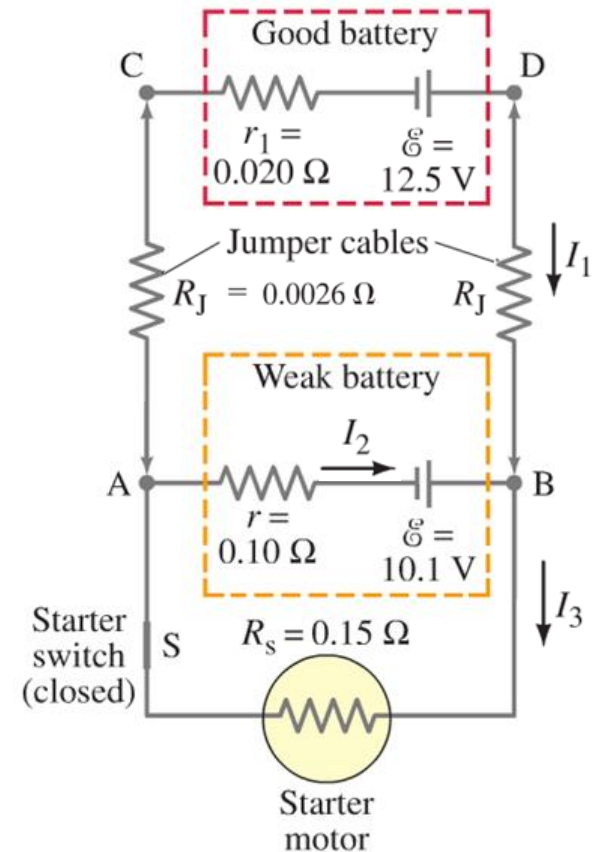
the lower loop

$$10.1 \text{ V} - I_3(0.15 \Omega) - I_2(0.10 \Omega) = 0$$

The junction rule at point B

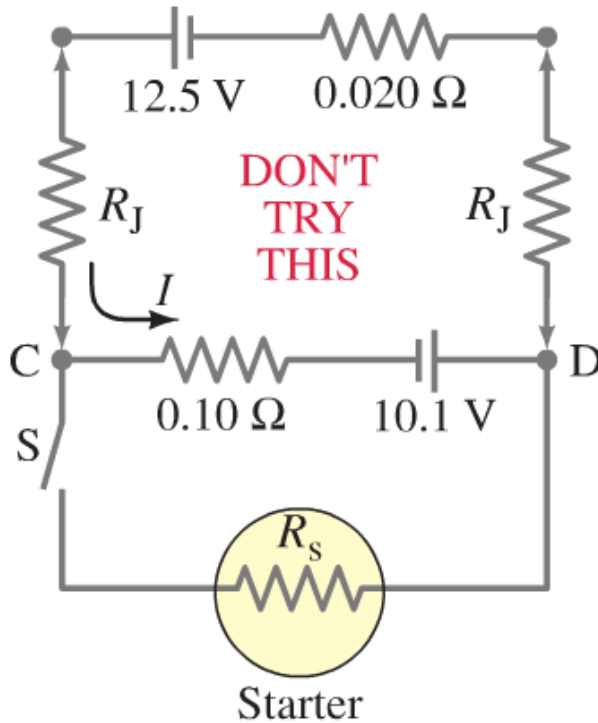
$$I_1 + I_2 = I_3$$

$$\begin{aligned} I_1 &= 76 \text{ A} \\ I_2 &= -5 \text{ A} \\ I_3 &= 71 \text{ A} \end{aligned}$$

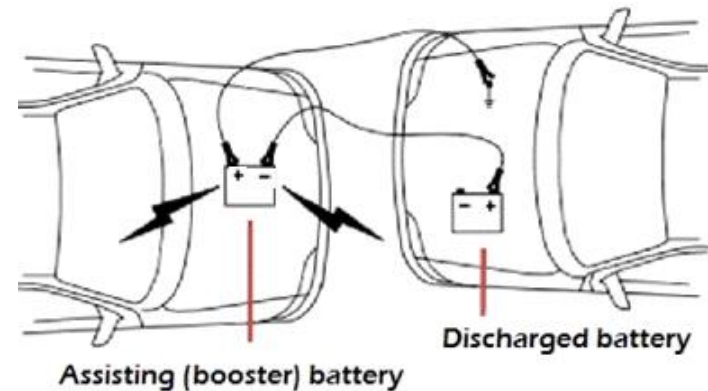


Jump starting a car

What if we connect it wrongly



if we connect it wrongly, i.e. + $\rightarrow$ -



Current $I = 180$ A before the motor is engaged !

Power dissipated in the weak battery

$$P = I^2 r = (180 \text{ A})^2 (0.10 \Omega) = 3200 \text{ W!}$$

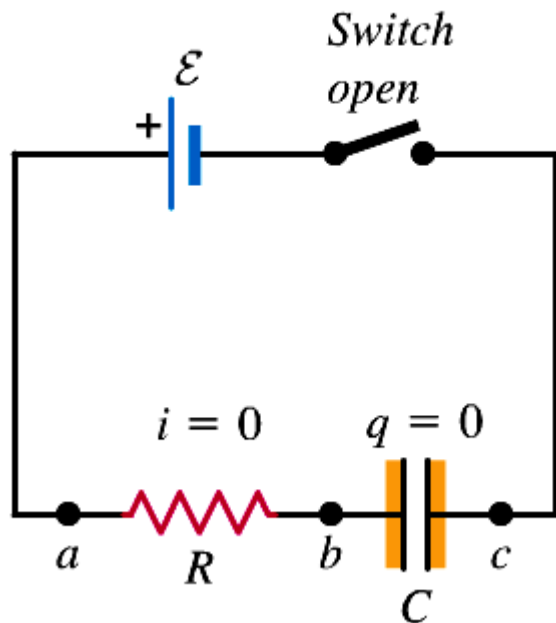
Today's topics (Chapter 27)

- Emf
- Kirchhoff Loop Rules
- RC Circuit

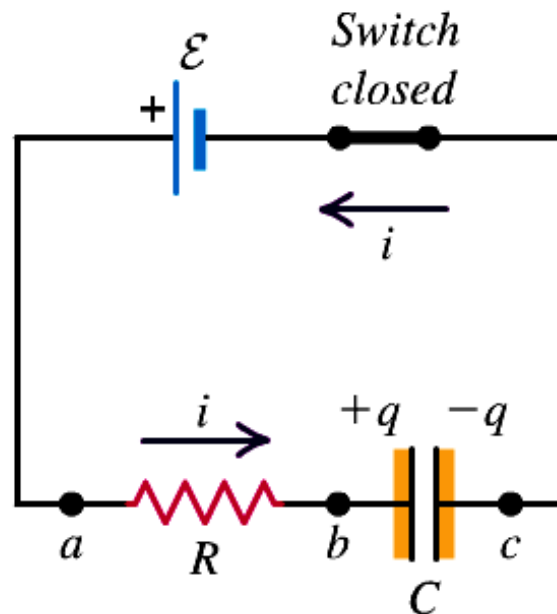
RC circuit

When charging or discharging a capacitor, the **currents (i)**, **voltages (v)** and **powers (p)** change with time

Capacitor initially uncharged



Charging the capacitor



When the switch is closed, the charge on the capacitor increases over time while the current decreases.

$$v_{ab} = iR \quad v_{bc} = \frac{q}{C}$$

By Kirchhoff's loop rule :

$$\mathcal{E} - iR - \frac{q}{C} = 0$$

RC circuit - charging the capacitor

$$\mathcal{E} - iR - \frac{q}{C} = 0$$

$$i = \frac{\mathcal{E}}{R} - \frac{q}{RC}$$

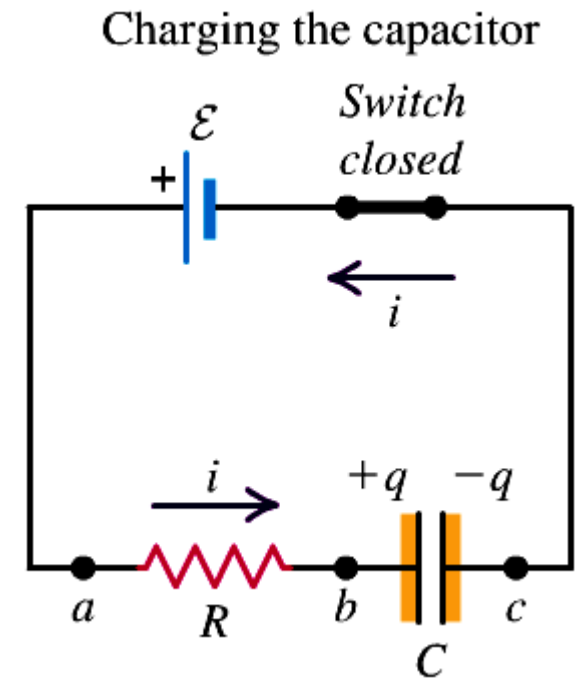
$$\frac{dq}{dt} = -\frac{1}{RC}(q - C\mathcal{E})$$

$$\frac{dq}{q - C\mathcal{E}} = -\frac{dt}{RC}$$

$$\int_0^q \frac{dq}{q - C\mathcal{E}} = -\int_0^t \frac{dt}{RC}$$

$$\ln\left(\frac{q - C\mathcal{E}}{-C\mathcal{E}}\right) = -\frac{t}{RC}$$

$$\frac{q - C\mathcal{E}}{-C\mathcal{E}} = e^{-t/RC}$$



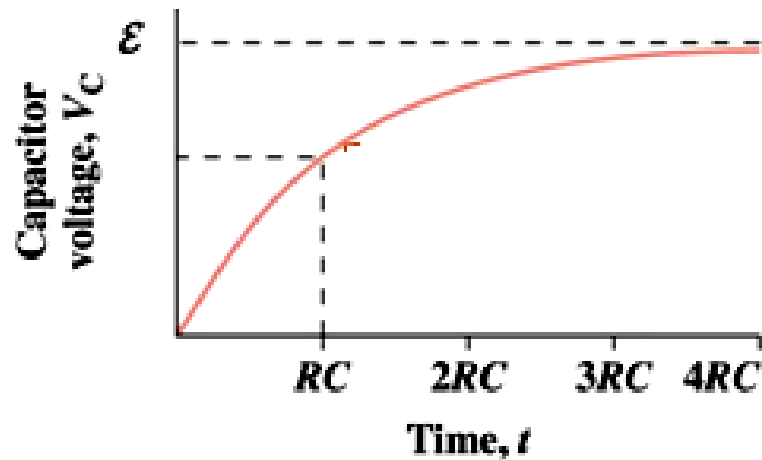
$$\therefore q = C\mathcal{E}(1 - e^{-t/RC})$$

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R}e^{-t/RC} = I_0 e^{-t/RC}$$

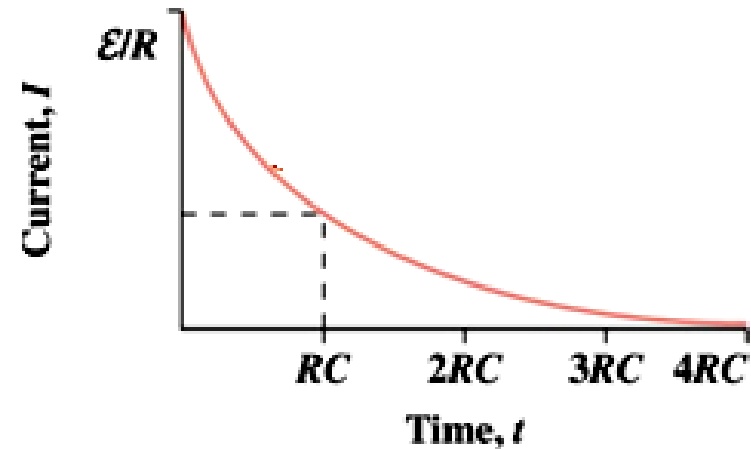
v (voltage across the capacitor) can be simply determined by $v = q/C$.

RC circuit – charging

$$V_C = \mathcal{E}(1 - e^{-t/RC})$$



$$i = I_0 e^{-t/RC}$$



At $t = 0$, $V_C = 0$

As $t \rightarrow \infty$, $V_C = \mathcal{E}$

Will the resistor affect the max charge stored in the capacitor?

RC circuit – time constant

$$q = C\mathcal{E}(1 - e^{-t/RC})$$

$$i = I_0 e^{-t/RC}$$

$$V_C = \mathcal{E}(1 - e^{-t/RC})$$

RC τ the time constant, the relaxation time
SI unit : second (s)

RC : the unit $\Omega \cdot \text{Farad}$ has the unit of time?

$$\begin{aligned} \frac{\text{Voltage}}{\text{Ampere}} \cdot \frac{\text{Coulomb}}{\text{Voltage}} &= \frac{\text{Coulomb}}{\text{Ampere}} \\ &= \frac{\text{time}}{\text{Coulomb}} \text{ Coulomb} \\ &= \text{time} \end{aligned}$$

RC circuit - discharging

$$\cancel{\mathcal{E}} - iR - \frac{q}{C} = 0$$

$$i = \frac{dq}{dt} = -\frac{q}{RC}$$

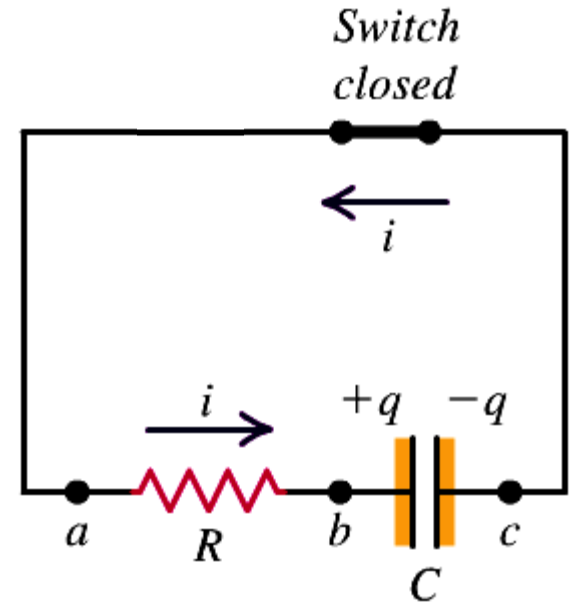
$$\int_{Q_0}^q \frac{dq}{q} = -\frac{1}{RC} \int_0^t dt$$

$$\ln \frac{q}{Q_0} = -\frac{t}{RC}$$

$$q = Q_0 e^{-t/RC}$$

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC} e^{-t/RC} = I_0 e^{-t/RC}$$

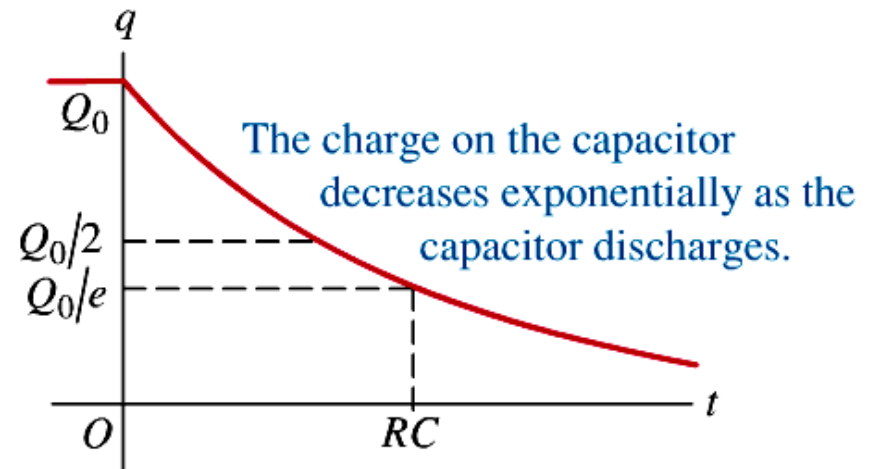
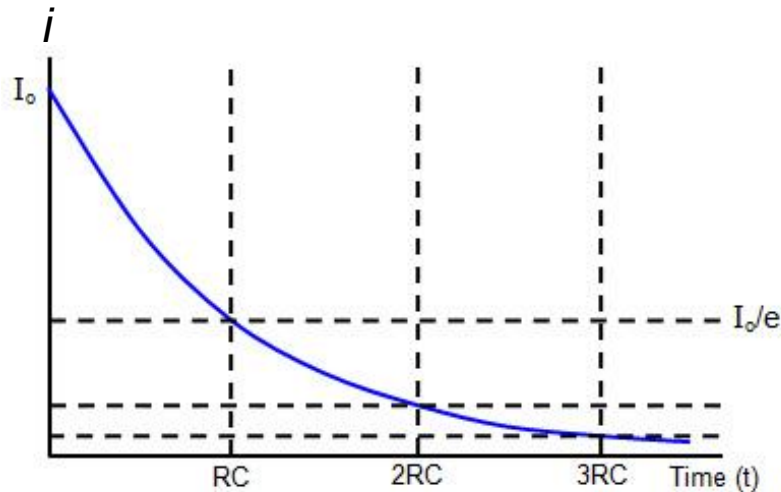
Discharging the capacitor



RC circuit - discharging

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0e^{-t/RC}$$

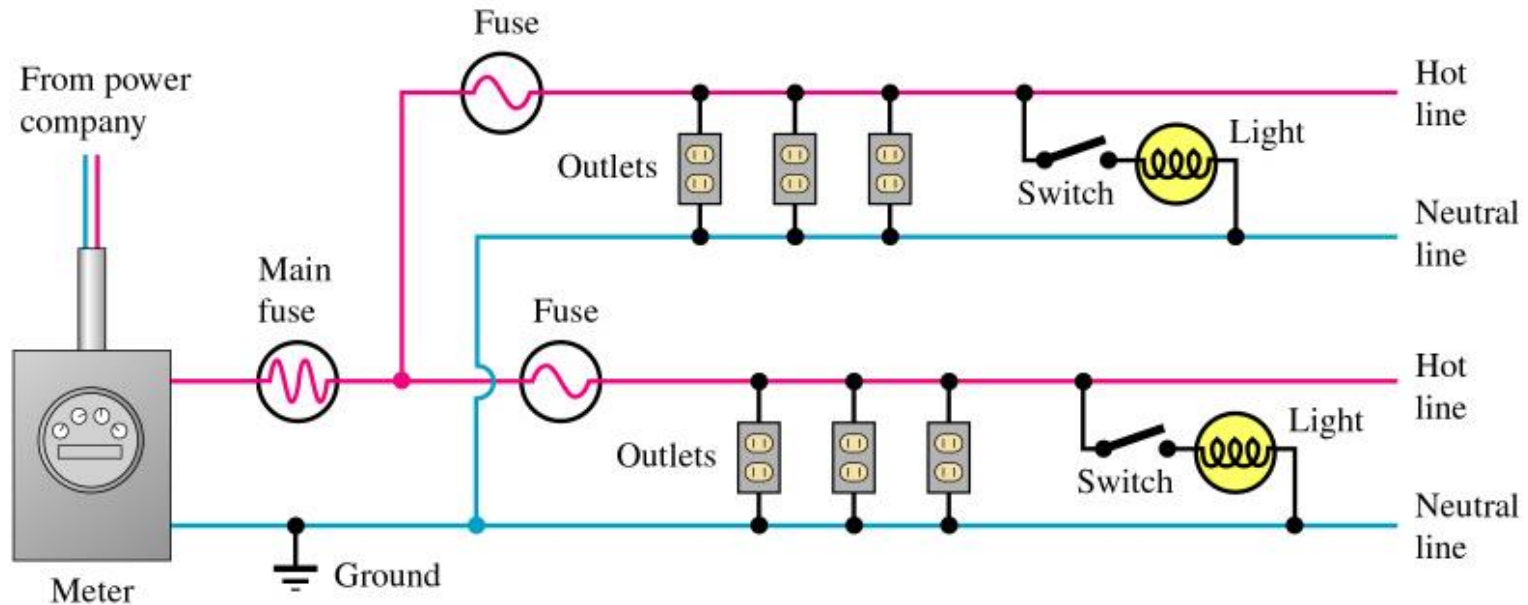
$$q = Q_0e^{-t/RC}$$



$$\mathcal{E} - iR - \frac{q}{C} = 0 \rightarrow \mathcal{E} = iR + \frac{q}{C}$$

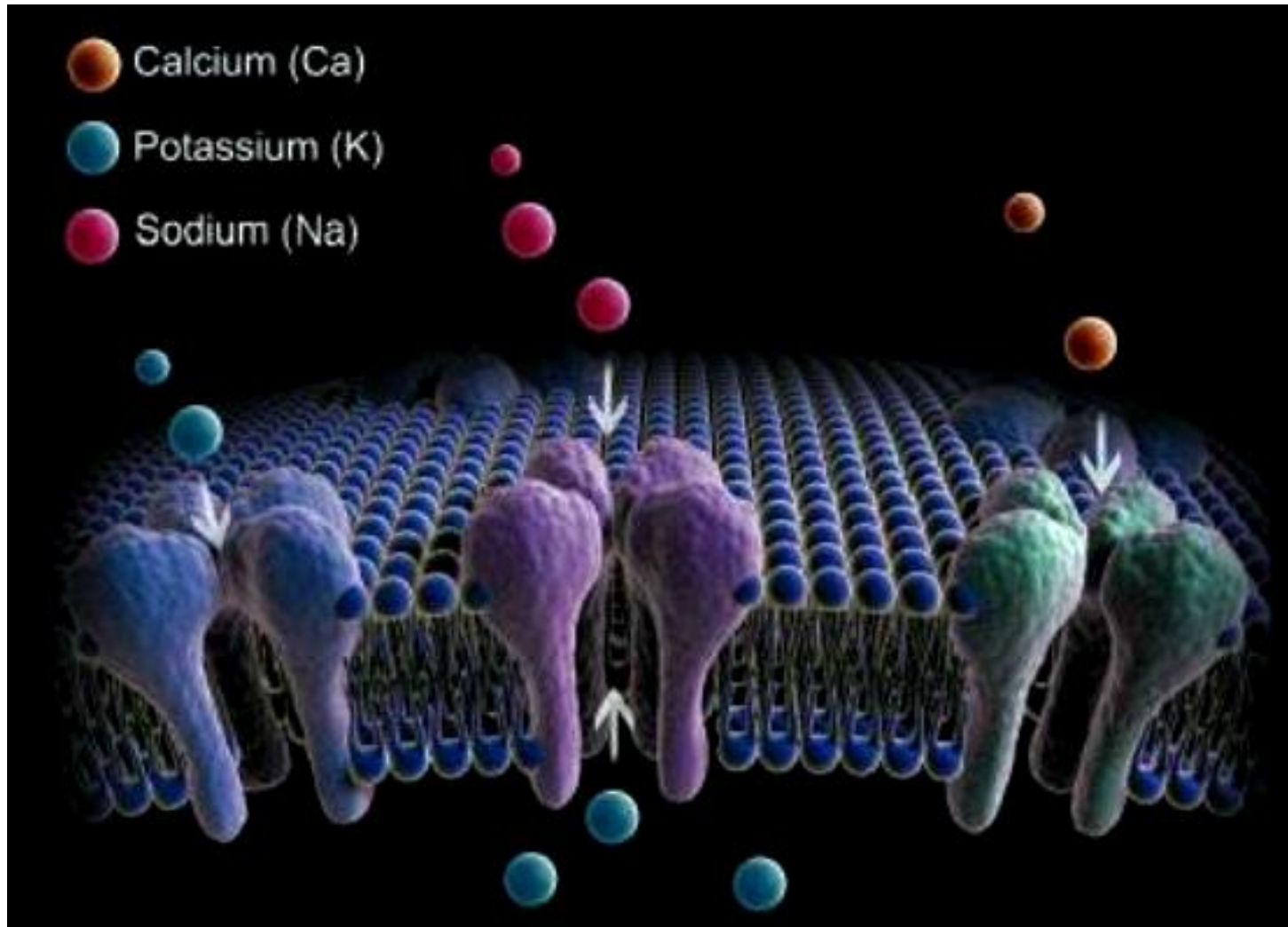
Energy supplied by the power : $\mathcal{E}i = i^2R + \frac{iq}{C}$

Power Distribution System



- Appliances are connected in parallel to power sources
- AC system in all household, commercial and industrial systems
- Fuse protects against overloading and overheating of circuits
- Fuse – lead-tin alloy with low melting temperature; melts when excess current flows
- Electrical energy is sold in kilowatt-hour (kWh), i.e. power x time

RC circuit in cell membrane

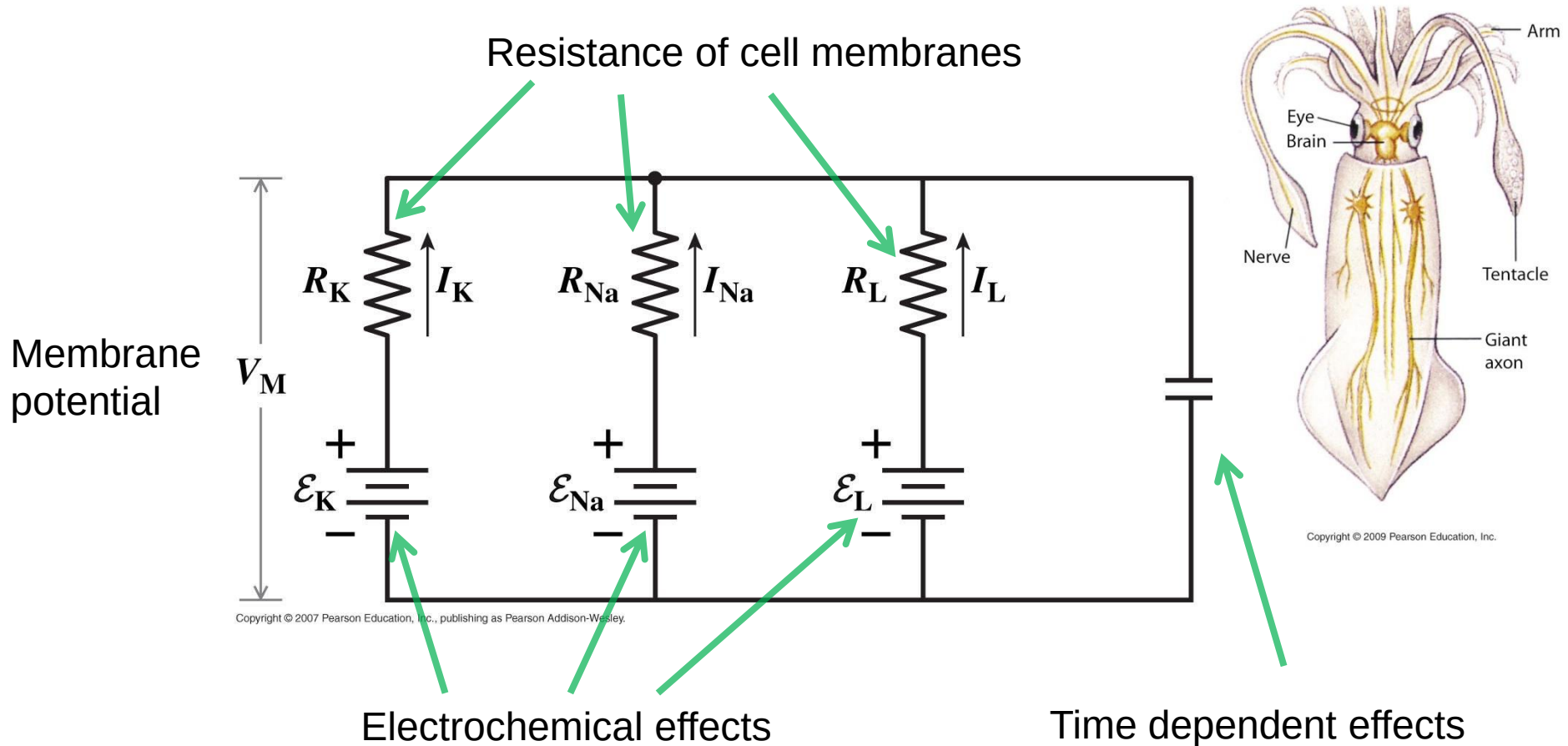


Circuit model of cell membrane

$$i = I_0 e^{-t/RC}$$

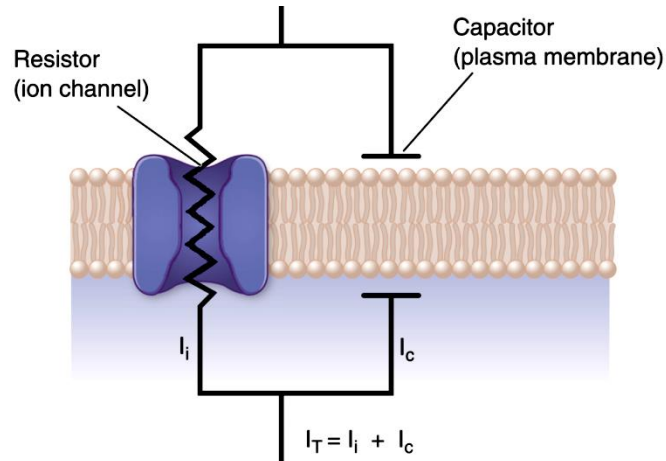
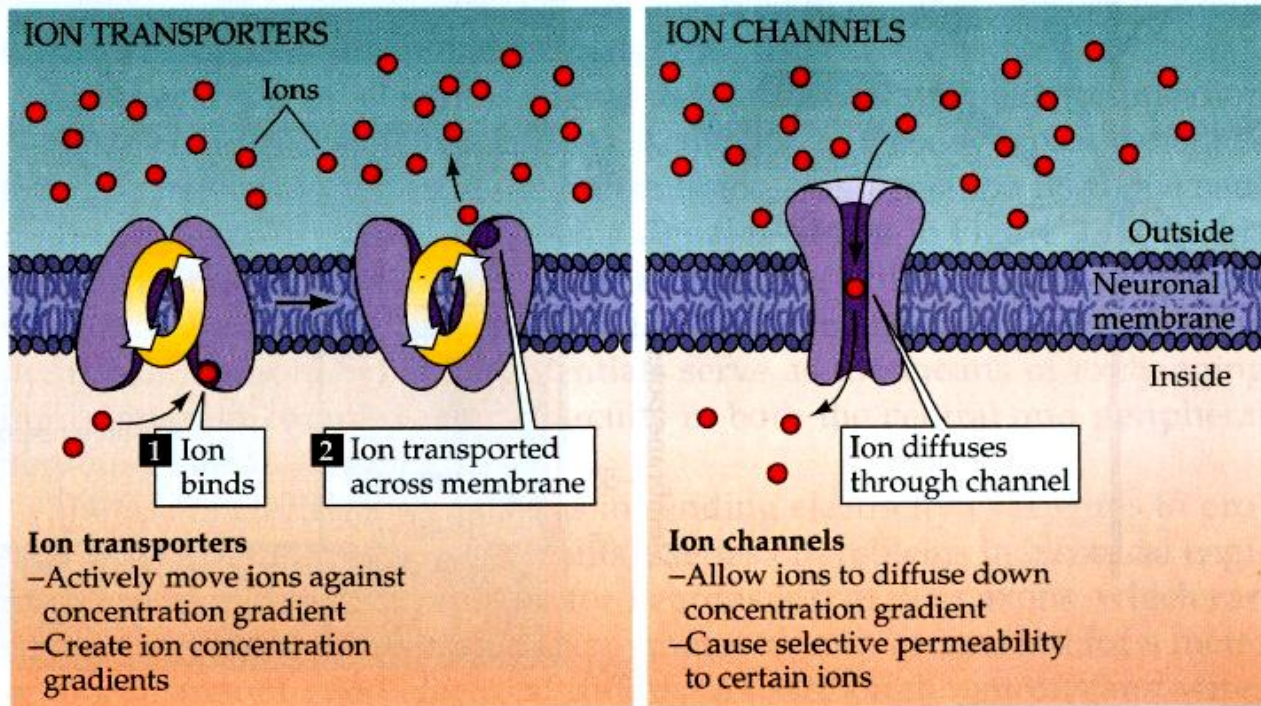
Charging or discharging a capacitor

Hodgkin-Huxley (1952) circuit model of cell membrane (Nobel prize, 1963):



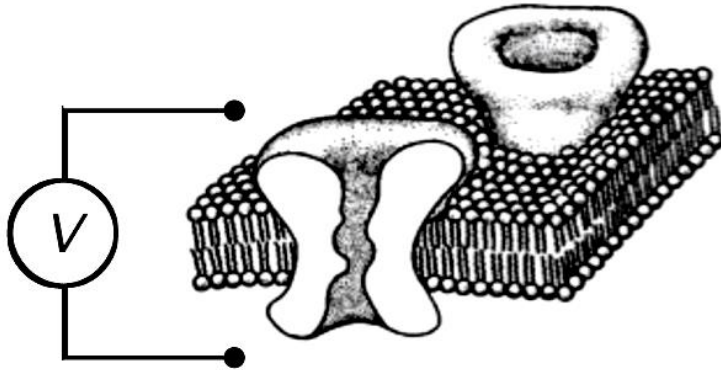
<https://zhuanlan.zhihu.com/p/102770891>

Ion motion across membrane



$$C \approx 1 \mu F / cm^2$$

Ion motion across membrane



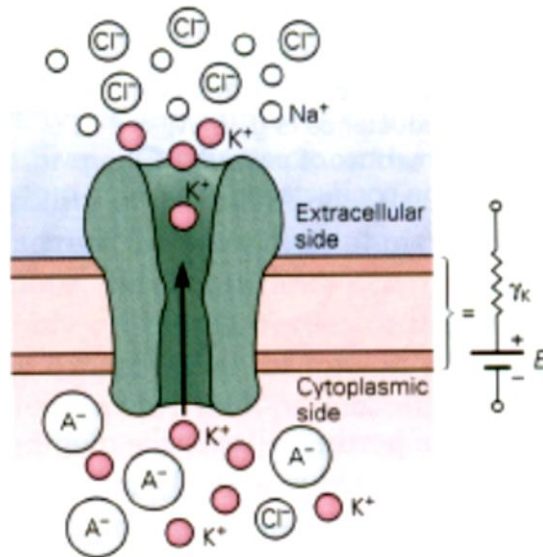
$$E_A = \frac{RT}{F} \ln \frac{[A]_o}{[A]_i}$$

R : universal gas constant
($JK^{-1}mol^{-1}$)

F : faraday constant
($C mol^{-1}$)

Extracellular and Intracellular Ion Concentrations

Ion	Concentration (mM)	
	Intracellular	Extracellular
Squid neuron		
Potassium (K^+)	400	20
Sodium (Na^+)	50	440
Chloride (Cl^-)	40-150	560
Calcium (Ca^{2+})	0.0001	10
Mammalian neuron		
Potassium (K^+)	140	5
Sodium (Na^+)	5-15	145
Chloride (Cl^-)	4-30	110
Calcium (Ca^{2+})	0.0001	1-2



$$E_K = -75 mV$$

$$E_{Na} = +55 mV$$

$$E_{Cl} = -70 mV$$

$$E_{Ca} \approx +150 mV$$

Example Camera Flash

A camera flash gets its energy from a 150- μF capacitor & requires 170 V to fire. If the capacitor is charged by a 200-V source through an 18-k Ω resistor, how long must the photographer wait between flashes? Assume the capacitor is fully charged at each flash.

Voltage during charging

$$V_C = \mathcal{E}(1 - e^{-t/RC})$$

$$\begin{aligned} t &= -RC \ln\left(1 - \frac{V_C}{\mathcal{E}}\right) \\ &= -(18 \times 10^3 \, \Omega)(150 \times 10^{-6} \, \text{F}) \ln\left(1 - \frac{170 \, \text{V}}{200 \, \text{V}}\right) \\ &= 5.1 \, \text{s} \end{aligned}$$

Summary

Review & Summary

Emf An **emf device** does work on charges to maintain a potential difference between its output terminals. If dW is the work the device does to force positive charge dq from the negative to the positive terminal, then the **emf** (work per unit charge) of the device is

$$\mathcal{E} = \frac{dW}{dq} \quad (\text{definition of } \mathcal{E}). \quad (27-1)$$

The volt is the SI unit of emf as well as of potential difference. An **ideal emf device** is one that lacks any internal resistance. The potential difference between its terminals is equal to the emf. A **real emf device** has internal resistance. The potential difference between its terminals is equal to the emf only if there is no current through the device.

Analyzing Circuits The change in potential in traversing a resistance R in the direction of the current is $-iR$; in the opposite direction it is $+iR$ (resistance rule). The change in potential in traversing an ideal emf device in the direction of the emf arrow is $+\mathcal{E}$; in the opposite direction it is $-\mathcal{E}$ (emf rule). Conservation of energy leads to the loop rule:

Loop Rule. *The algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero.*

Conservation of charge gives us the junction rule:

Junction Rule. *The sum of the currents entering any junction must be equal to the sum of the currents leaving that junction.*

Single-Loop Circuits The current in a single-loop circuit containing a single resistance R and an emf device with emf $\mathcal{E}$ and internal resistance r is

$$i = \frac{\mathcal{E}}{R + r}, \quad (27-4)$$

which reduces to $i = \mathcal{E}/R$ for an ideal emf device with $r = 0$.

Power When a real battery of emf $\mathcal{E}$ and internal resistance r does work on the charge carriers in a current i through the battery, the rate P of energy transfer to the charge carriers is

$$P = iV, \quad (27-14)$$

where V is the potential across the terminals of the battery. The rate P_r at which energy is dissipated as thermal energy in the battery is

$$P_r = i^2 r. \quad (27-16)$$

The rate P_{emf} at which the chemical energy in the battery changes is

$$P_{\text{emf}} = i\mathcal{E}. \quad (27-17)$$

Series Resistances When resistances are in **series**, they have the same current. The equivalent resistance that can replace a series combination of resistances is

$$R_{\text{eq}} = \sum_{j=1}^n R_j \quad (n \text{ resistances in series}). \quad (27-7)$$

Parallel Resistances When resistances are in **parallel**, they have the same potential difference. The equivalent resistance that can replace a parallel combination of resistances is given by

$$\frac{1}{R_{\text{eq}}} = \sum_{j=1}^n \frac{1}{R_j} \quad (n \text{ resistances in parallel}). \quad (27-24)$$

RC Circuits When an emf $\mathcal{E}$ is applied to a resistance R and capacitance C in series, as in Fig. 27-15 with the switch at a , the charge on the capacitor increases according to

$$q = C\mathcal{E}(1 - e^{-t/RC}) \quad (\text{charging a capacitor}), \quad (27-33)$$

in which $C\mathcal{E} = q_0$ is the equilibrium (final) charge and $RC = \tau$ is the **capacitive time constant** of the circuit. During the charging, the current is

$$i = \frac{dq}{dt} = \left(\frac{\mathcal{E}}{R}\right)e^{-t/RC} \quad (\text{charging a capacitor}). \quad (27-34)$$

When a capacitor discharges through a resistance R , the charge on the capacitor decays according to

$$q = q_0 e^{-t/RC} \quad (\text{discharging a capacitor}). \quad (27-39)$$

During the discharging, the current is

$$i = \frac{dq}{dt} = -\left(\frac{q_0}{RC}\right)e^{-t/RC} \quad (\text{discharging a capacitor}). \quad (27-40)$$

Brainstorm

Very simple question. We know the voltage between a and d is zero. But why there is a current?

